

Paul He

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Education

University of Toronto , BSc in Computer Science (Honours)	Sep 2021 – May 2025
- Graduated with High Distinction - Supervisors: Prof. Gerald Penn, Prof. Zhijing Jin	
ETH Zürich , Exchange Year in Computer Science	Sep 2023 – Aug 2024
Completed graduate-level coursework in NLP and Machine Learning	

Work Experience

Amazon Web Services (AWS) <i>Applied Scientist (Contract, Incoming)</i>	September 2025 – Present Tübingen, Germany
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Research Experience

Causal Reasoning Evaluation for Language Models <i>Supervisor: Prof. Zhijing Jin (Jinesis AI Lab)</i>	Jan 2025 – Aug 2025 Vector Institute
- Developed a tool to parse and evaluate causal expressions generated by LLMs, using BFS to match equivalent causal formulas and identify reasoning errors. - Discovered the ability to use this as a proxy to improve QA tasks involving causal understanding. - Submitted this work to EMNLP 2025.	
Polynomial-Time CCG Parsing: An Empirical Evaluation of the Kuhlmann–Satta Algorithm <i>Supervisor: Prof. Gerald Penn</i>	Sep 2024 - Aug 2025 University of Toronto
- Designed and implemented a polynomial-time parser for Combinatory Categorical Grammar (CCG). - Conducted large-scale multilingual experiments to analyze when symbolic parsers outperform naive baselines, identifying empirical complexity thresholds.	
Unsupervised Prompt Optimization for RAG using PMI <i>Supervisors: Tianyu Liu, Prof. Ryan Cotterell (Rycolab)</i>	Jun 2024 – Sep 2024 ETH Zürich
- Investigated attribution methods such as change in Jensen-Shannon divergence for Fisher information to approximate attention weights in LLMs. - Developed and implemented an unsupervised method to reorder retrieved documents using PMI, improving QA accuracy by up to 4% on NQ-Open and ELI5. - Presented the work at NAACL 2025 poster session, attracting sustained interest from researchers across academia and industry.	

Publications

CCG Revisited: A Multilingual Empirical Study of the Kuhlmann-Satta Algorithm <i>Paul He, Gerald Penn</i>	2025
<i>The 18th International Conference on Parsing Technologies</i> https://aclanthology.org/2025.iwpt-1.3/	
Pointwise Mutual Information as a Performance Gauge for Retrieval-Augmented Generation <i>Tianyu Liu, Jirui Qi, Paul He, Arianna Bisazza, Mrinmaya Sachan, Ryan Cotterell</i>	2025
<i>NAACL 2025 (Main)</i> , https://aclanthology.org/2025.naacl-long.78/	

Projects

Speeding up Earley’s Algorithm	Mar 2024 - Aug 2024
Proposed and implemented a grid search approach, which resulted in an algorithm $\times 5$ faster than the naive algorithm.	
Word Segmentation and Part-of-Speech Tagging with Transformer	Oct 2023 - Jan 2024
- Designed a new decoder embedding mechanism to better capture subtle linguistic nuances and dependencies. - Implemented the mechanism and achieved state-of-the-art results with a validation accuracy of 96.29%.	

Skills

Programming Languages: Python, C, Java, SQL

Spoken Languages: English (Fluent), Mandarin (Proficient), German (Proficient)